

DS-GAN: A Depthwise Separable GAN with Attention for Image Dehazing

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Abstract—Surveillance footage often suffers from haze and smoke, reducing visibility and hindering both human observers and automated systems. In this work, we present DS-GAN, a deep learning model that employs depthwise separable convolutions and squeeze-and-excitation (SE) blocks to efficiently remove haze from images. The model is trained and evaluated using a well-known publicly available RESIDE OTS synthetic dataset. Performance is assessed with standard image quality metrics, including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). Experimental results show that DS-GAN achieves highly competitive dehazing quality with a PSNR of 35.08 dB and SSIM of 0.9722. Beyond surveillance applications, this efficient approach enables deployment in autonomous vehicles for fog removal, robotics for clarity enhancement, and any computer vision application where atmospheric disturbances compromise visibility.

Index Terms—Image dehazing, smoke removal, GAN, depthwise separable convolution, SE block

I. INTRODUCTION

Haze and smoke present significant obstacles in surveillance footage, often obscuring people and objects and making critical monitoring tasks more difficult. This degradation is especially problematic during emergencies such as fires, where reduced visibility can delay the identification and rescue of individuals. In such scenarios, both white and black smoke—formed by airborne particles scattering light—can either partially or completely block important details in video streams.

To address this challenge, researchers have turned to image dehazing methods that aim to restore clarity in degraded scenes. Recent advances in synthetic datasets, such as RESIDE OTS, allow for systematic evaluation of these algorithms by providing pairs of hazy and clear images. Improving image quality, even modestly, can greatly assist emergency responders and automated systems tasked with detection and tracking in adverse conditions. In this context, the development of efficient deep learning techniques for real-time dehazing is of growing importance. Figure 1 provides a visual example of the typical input and target pairs used for training and evaluation throughout this work. The ability to transform degraded images (left) into visually clear



Fig. 1: An example of (a) Hazy image and (b) Ground truth of the hazy image

counterparts (right) is the main objective behind this study.

In recent years, deep learning has revolutionized image dehazing, leading to a surge in neural network-based approaches that map hazy images directly to their clear versions. Most existing methods use convolutional neural networks (CNNs) to extract features and restore visual clarity. However, their performance can be limited by the size and diversity of available training data [1]. Further, the use of a complex deep learning architecture not only increases the training time and energy consumption but also increases the number of learnable parameters, which eventually increases the memory requirements and hence not suitable for real-time applications. This study aims to develop a robust and efficient model for smoke and haze removal by leveraging generative adversarial networks (GANs), with a focus on minimizing computational demands. We introduce an enhanced GAN architecture that integrates depthwise separable convolutions, modular residual blocks, and squeeze-and-excitation (SE) attention, which together improve feature representation while keeping the model lightweight.

II. LITERATURE REVIEW

Reducing risks to human safety in fire and smoke-filled environments remains a major challenge for automated vision systems. The presence of haze and smoke in surveillance footage can result in severe visibility loss, often complicating rescue efforts and impeding situational awareness. Recognizing the simi-

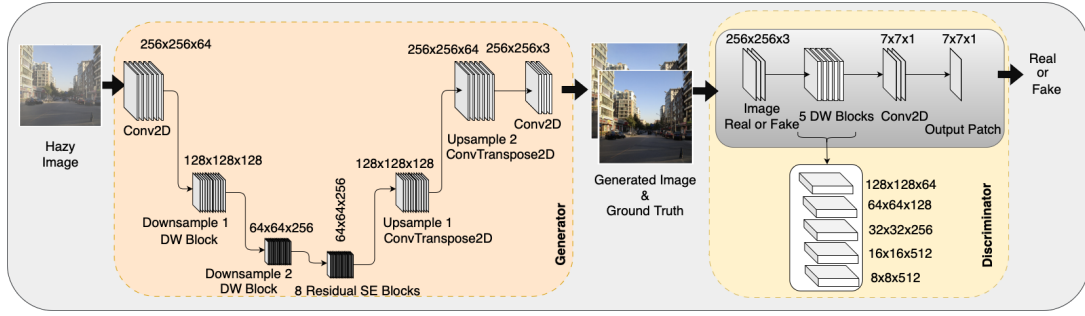


Fig. 2: Architecture of the proposed DS-GAN

larity between haze and smoke distortion, many methods developed for dehazing have been adapted to address desmoking tasks.

Early deep learning approaches such as DehazeNet [2], AOD-Net [3], and MSCNN [4] demonstrated that CNN-based models could outperform traditional image processing in both speed and quality. Later, attention mechanisms and multi-scale designs, as in GridDehazeNet, FFA-Net, MSBDN, and AE-CRNet [5], [6], produced further gains and have set new benchmarks in the field. Generative adversarial networks (GANs) have been especially impactful for perceptual quality: Fu et al. [1] proposed DW-GAN, which combines discrete wavelet transforms with GANs for dehazing and desmoking; Gkika et al. [7] demonstrated DW-GAN’s utility for object detection in smoke; Huang and Chen [8] developed an Attentive GAN for desmoking, and Bharath Raj and Venkateswaran [9] used a U-Net GAN for haze removal. Other works have targeted real-time surveillance [10], surgical video [11], and industrial inspection [12].

The most recent literature sees the rise of transformer-based models and hybrid methods that blend physical image priors with deep networks, such as TCFDN [13] and PhysicsGAN [14]. Variational approaches like EVID [15] and cooperative networks for medical imaging [11] further illustrate the diversity of strategies being explored for robust haze and smoke removal. The rest of the paper is organized as follows: Section III discusses the methodology used to perform experiments. The results are presented and discussed in Section IV. Finally, the Section V concludes the study.

III. METHODOLOGY

The proposed framework aims to restore visibility in images degraded by haze or smoke, supporting downstream tasks such as detection and tracking in adverse environments. The proposed approach focuses on an enhanced generative adversarial network (GAN) that uses depthwise separable convolutions, modular residual blocks, and squeeze-and-excitation (SE) channel attention to efficiently capture and recover fine image structure while minimizing model complexity.

A. Dataset and Preprocessing

We use the Outdoor Training Set (OTS) from the Realistic Single Image Dehazing (RESIDE) dataset [16], containing over 72,000 synthetic hazy images paired with clear ground truths. Hazy and clear images are stored in separate directories and paired by matching base filenames (e.g., hazy123.jpg with 123.jpg). All images are resized to 256×256 pixels. During preprocessing, images are normalized to $[-1, 1]$ using a fixed mean of 0.5 and standard deviation of 0.5 per channel, which centers and scales pixel values without performing zero-mean/unit-variance normalization. Data augmentation is limited to random horizontal flipping during training. The dataset is partitioned randomly into 80% for training and 20% for validation.

B. DS-GAN Architecture

The proposed DS-GAN model (as shown in Figure 2) replaces standard convolutions with depthwise separable and pointwise convolutions throughout both the generator and discriminator. Modular residual blocks and SE channel attention further improve feature learning and model compactness. The following describes the network components:

1) *Depthwise Separable Convolutions*: In a typical convolutional layer, spatial filtering and channel mixing are performed jointly, requiring $D_k \times D_k \times C_{in} \times C_{out}$ parameters (where D_k is kernel size, C_{in} input channels, C_{out} output channels). As illustrated in Figure 3, Depthwise Separable Convolution is a type of convolution that reduces computation by factorizing standard convolution into two steps:

- **Depthwise convolution**: Applies a spatial filter to each channel independently, using $D_k \times D_k \times C_{in}$ parameters.
- **Pointwise convolution**: Follows with a 1×1 convolution that mixes information across channels, using $C_{in} \times C_{out}$ parameters.

This separation significantly reduces the number of parameters and computational cost compared to standard convolution. The final output has the same dimensions as a regular convolution but is much more efficient. This

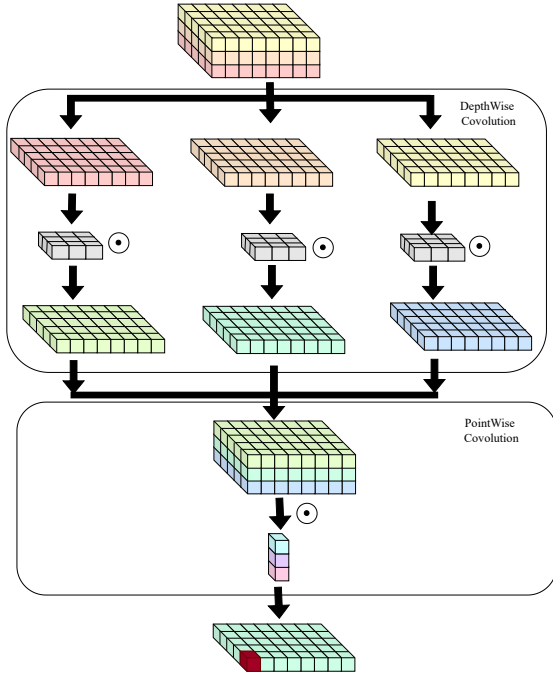


Fig. 3: Architecture of the Depthwise separable and pointwise convolutional operation

technique is widely used in lightweight neural networks such as MobileNet. It enables real-time performance on resource-constrained devices like smartphones or embedded systems [17], [18].

2) *Squeeze-and-Excitation (SE) Blocks*: SE blocks are integrated within residual blocks to provide adaptive channel-wise recalibration. Each SE block applies global average pooling to compress spatial information, then uses two fully connected layers and a sigmoid activation to generate per-channel weights. These weights rescale the feature maps, enabling the network to focus on the most informative channels for dehazing.

3) *Generator*: The generator follows a U-Net inspired encoder-bottleneck-decoder design with skip connections:

- **Input**: 7×7 convolution (stride 1, padding 3) expands 3 channels to 64, followed by instance normalization and ReLU.
- **Encoder**: Two depthwise separable blocks with stride 2 ($64 \rightarrow 128$, $128 \rightarrow 256$), each with instance normalization and LeakyReLU ($\alpha = 0.2$).
- **Bottleneck**: Six residual blocks, each combining DWBlocks with SE-based channel attention, stabilized by residual connections.
- **Decoder**: Two transposed convolutions for upsampling ($256 \rightarrow 128$, $128 \rightarrow 64$) with skip connections, instance normalization, and ReLU.
- **Output**: 7×7 convolution maps features back to 3 channels, followed by Tanh for $[-1, 1]$ scaling.

4) *Discriminator*: The discriminator adopts a PatchGAN architecture, which evaluates the realism of local patches instead of assigning a single global score. Specifically, the model outputs a $7 \times 7 \times 1$ patch map, where each element corresponds to the discriminator's confidence that a local 70×70 region of the input image is real. This patch-level evaluation enforces local realism, compelling the generator to refine fine-grained textures and edge details.

The architecture consists of an initial convolutional layer followed by five depthwise separable convolution (DW) blocks with instance normalization, LeakyReLU activations, and dropout for regularization. This lightweight design reduces parameters while maintaining strong discriminative capacity.

a) *PatchGAN Fundamentals*: Patch-based discrimination is particularly effective for image-to-image translation tasks such as dehazing, as it emphasizes spatial consistency and enhances perceptual sharpness. By encouraging realism at the patch level, the discriminator prevents overly smooth reconstructions and promotes high-frequency detail preservation, leading to cleaner and more visually convincing outputs.

C. Training Configuration and Hyperparameters

All experiments were implemented using PyTorch on an NVIDIA RTX 6000 Ada GPU. The training process employs a multi-objective loss function and carefully tuned hyperparameters to ensure stable GAN training and optimal convergence.

a) *Optimization Strategy*: To address the inherent training instability in GANs, we employ differentiated learning rates for the generator and discriminator. The generator uses Adam optimizer with learning rate $lr_G = 1 \times 10^{-4}$, while the discriminator uses $lr_D = 4 \times 10^{-4}$ to ensure balanced adversarial training. Both optimizers use momentum parameters $\beta_1 = 0.5$ and $\beta_2 = 0.999$. Cosine annealing learning rate scheduling is applied with $T_{max} = \text{num_epochs}$ to promote smooth convergence.

b) *Training Stabilization Techniques*: Several techniques are incorporated to enhance training stability:

- **Label smoothing**: Real labels are set to 0.9 and fake labels to 0.1 to prevent discriminator overconfidence.
- **Gradient clipping**: Maximum gradient norm is limited to 1.0 for both networks to prevent gradient explosion.
- **Asymmetric training**: The generator is updated every second iteration while the discriminator is updated every iteration, maintaining training balance.
- **Instance normalization**: Used throughout both networks instead of batch normalization for better

training stability in image-to-image translation tasks.

c) Network Initialization and Regularization:

All convolutional layers are initialized using normal distribution with mean 0.0 and standard deviation 0.02. Batch normalization layers use normal initialization (mean 1.0, std 0.02) for weights and zero initialization for biases. Dropout with probability 0.3 is applied in discriminator's depthwise separable blocks for regularization.

d) Training Configuration: The models are trained for 35 epochs with batch size 4, constrained by GPU memory limitations. Early convergence was achieved due to the stabilization techniques employed. The SE blocks use a reduction ratio of 16 for channel attention mechanisms. All input images are resized to 256×256 pixels and normalized to the range $[-1, 1]$ using mean 0.5 and standard deviation 0.5 per channel. Data augmentation is limited to random horizontal flipping during training to preserve the natural characteristics of haze distribution.

e) Inference Optimization: For deployment efficiency, several inference optimizations are applied post-training:

- **Half precision (FP16):** Reduces memory usage by approximately 50% and improves inference speed on compatible hardware.
- **TorchScript compilation:** Enables graph optimization and removes Python overhead during inference.
- **CUDA optimizations:** Enables cuDNN benchmarking and optimizes memory allocation patterns.

D. Training and Evaluation

The DS-GAN model is trained on the RESIDE OTS dataset using the preprocessing and splits described earlier. The training objective combines adversarial learning with pixel-wise reconstruction and perceptual consistency:

$$\mathcal{L}_{total} = \mathcal{L}_{GAN} + \lambda_{L1}\mathcal{L}_{L1} + \lambda_{perc}\mathcal{L}_{perc} \quad (1)$$

where:

- $\mathcal{L}_{GAN}$ is the adversarial term, implemented as a binary cross-entropy loss (BCEWithLogits) to encourage photo-realistic outputs.
- $\mathcal{L}_{L1}$ is the pixel-level reconstruction loss, enforcing similarity between the generated and ground-truth images.
- $\mathcal{L}_{perc}$ is a perceptual loss, computed as mean squared error in the image space or feature space, promoting structural fidelity.
- λ_{L1} and λ_{perc} balance the contribution of each term, where $\lambda_{L1} = 10$ and $\lambda_{perc} = 0.1$.

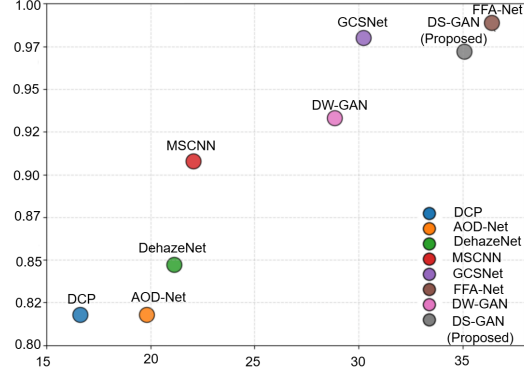


Fig. 4: Comparison of dehazing methods in PSNR and SSIM. Marker size reflects SSIM. DS-GAN balances high accuracy with efficiency.

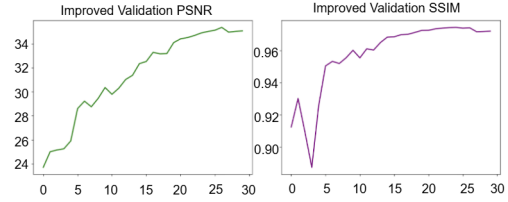


Fig. 5: Distributions of PSNR and SSIM across the OTS validation set. DS-GAN shows stable performance.

To improve stability, we apply label smoothing for the discriminator, gradient clipping, and Adam optimizers with separate learning rates for the generator and discriminator. Cosine annealing schedulers further regulate learning dynamics.

E. Evaluation Metrics

Performance is measured using two standard metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

PSNR: Quantifies reconstruction fidelity as

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right), \quad (2)$$

where MAX_I is the maximum pixel intensity and MSE is the mean squared error. Higher values indicate closer pixel-wise agreement.

SSIM: Captures perceptual quality by comparing luminance, contrast, and structure:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}. \quad (3)$$

Values range from 0 (no similarity) to 1 (perfect match). In practice, scores above 0.8 are considered visually convincing for restoration tasks.

IV. RESULTS AND ANALYSIS

This study evaluates the performance of our proposed Depthwise Separable GAN (DS-GAN) model for image

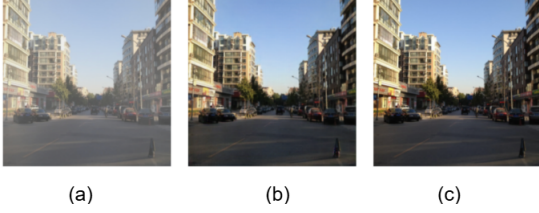


Fig. 6: Qualitative comparison: (a) hazy input, (b) DS-GAN output and (c) Ground truth.

TABLE I: Performance comparison of image dehazing methods on RESIDE OTS.

Method	Authors	PSNR (dB)	SSIM
DCP	He et al.	16.62	0.8179
AOD-Net	Li et al.	19.82	0.8178
DehazeNet	Cai et al.	21.14	0.8472
MSCNN	Ren et al.	22.06	0.9078
GCSNet	Chen et al.	30.23	0.9800
FFA-Net	Qin et al.	36.39	0.9890
DW-GAN	Fu et al.	28.83	0.9330
DS-GAN (Ours)	Proposed	35.08	0.9722

TABLE II: Ablation: effect of depthwise separable convolutions.

Model	PSNR (dB)	SSIM
Standard Conv Baseline	22.89	0.8421
DS-GAN (Ours)	35.08	0.9722

dehazing on the widely-used RESIDE OTS dataset. The experiments were conducted using an NVIDIA RTX 6000 Ada GPU. We employed PyTorch for model implementation and training. The model was evaluated using standard image quality metrics including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

A. Quantitative Results

DS-GAN achieves an average PSNR of 35.08 dB and SSIM of 0.9722, as summarized in Table I. While FFA-Net and GCSNet obtain slightly higher absolute values, our model delivers competitive restoration quality with far fewer parameters, highlighting its efficiency–accuracy balance.

Figure 4 illustrates the PSNR–SSIM tradeoff. DS-GAN clusters among top-performing methods while maintaining compactness, making it suitable for real-time or resource-limited deployment. Across the validation set, standard deviations of 2.67 dB (PSNR) and 0.0166 (SSIM) confirm consistent performance across diverse haze levels and scenes.

B. Qualitative Results

Figure 6 shows that DS-GAN restores visibility while preserving color fidelity and structural detail. Figure 5 further demonstrates tight distributions of PSNR and SSIM, underlining the model’s robustness. Key strengths include:

TABLE III: DS-GAN performance across haze densities.

Haze Density	PSNR (dB)	SSIM
Light	36.20	0.9835
Medium	35.08	0.9722
Heavy	33.72	0.9583

- 1) **Color restoration:** The model effectively recovers natural colors that were distorted by haze, avoiding color shift issues that sometimes occur with physics-based methods like DCP.
- 2) **Detail preservation:** Fine structures and textures are well-preserved in the dehazed images, which is particularly noticeable in complex scenes with multiple objects.
- 3) **Contrast enhancement:** The restored images show improved contrast compared to the hazy inputs, enhancing overall visibility without introducing artificial artifacts.
- 4) **Edge sharpness:** Object boundaries are well-defined in the dehazed results, indicating that the model maintains structural integrity during the dehazing process.

C. Ablation Study and Robustness

To understand the individual contributions of our design choices, we conducted systematic ablation experiments focusing on the core architectural components of DS-GAN.

1) *Impact of Depthwise Separable Convolutions:* We investigated the effectiveness of depthwise separable convolutions by creating a baseline variant where all depthwise separable layers were replaced with standard convolutions, while keeping other components identical. As shown in Table II, this architectural change resulted in a dramatic performance decline. The baseline achieved only 22.89 dB PSNR and 0.8421 SSIM, representing substantial drops of 12.19 dB and 0.1301 respectively compared to our full DS-GAN model.

This significant degradation highlights two key insights: first, that depthwise separable convolutions are not merely parameter-reduction techniques but genuinely improve feature learning for dehazing tasks. Second, the factorization of spatial and channel-wise operations appears particularly well-suited to the complex patterns present in hazy imagery, where atmospheric scattering creates intricate spatial-spectral relationships.

2) *Performance Across Varying Haze Conditions:* Real-world surveillance environments present diverse atmospheric conditions, making robustness across different haze intensities crucial for practical deployment. We analyzed DS-GAN’s performance by categorizing validation images into light, medium, and heavy haze groups based on visibility degradation patterns.

As demonstrated in Table III, DS-GAN exhibits consistent restoration capabilities across this spectrum.

While performance naturally decreases with increasing haze density, the degradation is gradual rather than catastrophic. The model maintains above 33 dB PSNR even for heavily degraded images, with SSIM values consistently exceeding 0.95 across all conditions.

Particularly noteworthy is the relatively small performance gap between light and heavy haze scenarios (2.48 dB PSNR difference), suggesting that our SE-enhanced residual blocks effectively adapt to varying atmospheric conditions. This robustness stems from the attention mechanism's ability to emphasize relevant channel features regardless of haze intensity, making DS-GAN suitable for deployment in unpredictable environmental conditions typical of surveillance applications.

The consistent performance across haze densities validates our architectural choices and demonstrates the model's practical viability for real-world monitoring systems where atmospheric conditions cannot be controlled.

V. CONCLUSION

We introduced DS-GAN, a GAN-based dehazing model that integrates depthwise separable convolutions and squeeze-and-excitation blocks. Experiments on RESIDE OTS show that DS-GAN effectively dehazes images and achieves a high PSNR & SSIM while remaining compact, outperforming conventional baselines and approaching state-of-the-art methods. Robustness across haze densities and efficiency make it promising for deployment in real-world vision systems.

Extending this study, in future, we will work on increasing the video frames per second to make it efficient for real-time applications. We also plan to evaluate DS-GAN on real-world surveillance footage and investigate its integration with downstream computer vision tasks such as object detection and tracking in adverse weather conditions.

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